Automated Keratoconus Screening With Corneal Topography Analysis

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Purpose. Although visual inspection of corneal topography maps by trained experts can be powerful, this method is inherently subjective. Quantitative classification methods that can detect and classify abnormal topographic patterns would be useful. An automated system was developed to differentiate keratoconus patterns from other conditions using computer-assisted videokeratoscopy.

Methods. This system combined a classification tree with a linear discriminant function derived from discriminant analysis of eight indices obtained from TMS-1 videokeratoscope data. One hundred corneas with a variety of diagnoses (keratoconus, normal, keratoplasty, epikeratophakia, excimer laser photorefractive keratectomy, radial keratotomy, contact lens-induced warpage, and others) were used for training, and a validation set of 100 additional corneas was used to evaluate the results.

Results. In the training set, all 22 cases of clinically diagnosed keratoconus were detected with three false-positive cases (sensitivity 100%, specificity 96%, and accuracy 97%). With the validation set, 25 out of 28 keratoconus cases were detected with one false-positive case, which was a transplanted cornea (sensitivity 89%, specificity 99%, and accuracy 96%).

Conclusions. This system can be used as a screening procedure to distinguish clinical keratoconus from other corneal topographies. This quantitative classification method may also aid in refining the clinical interpretation of topographic maps. Invest Ophthalmol Vis Sci. 1994;35:2749–2757.

Keratoconus is one of the noninflammatory corneal thinning disorders characterized by anterior protrusion of the cornea and stromal thinning.[1] Detection of keratoconus is important to avoid unpredictable results in refractive surgery and difficulties in contact lens fitting. Additionally, screening for the early form of keratoconus will lead to a better understanding of the natural course of keratoconus, as well as the genetic basis of this disease.

Although advanced keratoconus is easily diagnosed by slit lamp findings and keratometry readings, it is difficult to detect early keratoconus with these instruments.[2] Photokeratoscopy[1,3] and computer-assisted videokeratography[2,4,5] allow early keratoconus to be detected by the trained observer. In particular, computer-assisted videokeratoscopy has been used for analyzing irregular astigmatism seen in the family members of patients with keratoconus,[6,7] following up the progression of subclinical keratoconus,[8] and assisting in contact lens selection for keratoconus.[9]

Interpretation of videokeratographs requires examiners to have prior training in discerning the complex patterns or subtle features contained in the contour map. Topographic maps of eyes with keratoconus display a variety of different patterns[5] that may be misinterpreted by an untrained examiner. Therefore, automatic quantitative image analysis of color-coded maps and objective criteria for interpreting a true keratoconus pattern would be useful.

Although numerical methods have been developed to distinguish the keratoconus cornea from the normal cornea,[4] there have been no reports of a method to differentiate between keratoconus and a number of other clinical entities whose topographic presentation may share features similar to those of

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keratoconus. Therefore, we used eight quantitative indices derived from videokeratoscopy data (five new, three preexisting) as input to an automated keratoconus pattern detection algorithm that uses discriminant analysis embedded in an artificial intelligence mesh.

METHODS

Subjects

Color-coded maps and data files generated by a computer-assisted videokeratoscope (TMS-1, Computed Anatomy, NY) were used. Institutional Review Board approval was obtained, all subjects signed informed consent forms, and the tenets of the Declaration of Helsinki were followed in taking videokeratographs.

Maps were categorized by three corneal topography researchers based on clinical records and topographic appearances with the fixed Klyce/Wilson scale (28.0 to 65.5 D in 1.5 D steps). Videokeratographs were drawn from the Louisiana State University Eye Center patient population and were divided randomly by category into two sets. Corneas with mixed diagnoses, as indicated in the medical records, and poorly centered videokeratographs were excluded. The keratoconus detection program was developed using a training set of 100 corneas and evaluated with a validation set of an additional 100 corneas (Table 1). Each set comprised eight categories: normal, keratoconus, keratoplasty, epikeratophakia, excimer laser photorefractive keratectomy, radial keratotomy, contact lens-induced warpage, and other. The distribution of diagnoses was similar in the two sets (Table 1). The normal category included corneas with regular astigmatism ranging from 1.5 D to 3.75 D (seven maps in the training set and nine maps in the validation set). The keratoconus category encompassed a range of severity from mild to advanced clinically diagnosed keratoconus; 11 of the 50 eyes had centrally located cones. Corneas with localized areas of steepness on videokeratography without conventional clinical findings (keratoconus suspect) were excluded. The category marked other consisted of 12 eyes in the training set (four relaxing incisions, four scarred corneas, two post-retinal detachment surgery, one post-cataract surgery, and one keratomileusis) and eight eyes in the validation set (seven scarred corneas and one pterygium).

TABLE 1. Subjects in Each Set

	Training Set	*Validation Set*
Keratoconus	22	28
Mild	(7)	(8)
Moderate	(7)	(10)
Advanced	(8)	(10)
Normal cornea	30	30
Regular astigmatism	(7)	(9)
Keratoplasty	15	14
Epikeratophakia	7	7
Photorefractive keratectomy	5	4
Radial keratotomy	5	4
Contact lens-induced warpage	4	5
Other	12	8
Total	100 Eyes	100 Eyes

Quantitative Indices of the Topographic Map

The statistical indices calculated for every topographic map included three existing indices—Simulated K1 (SimK1), Simulated K2 (SimK2), and Surface Asymmetry Index (SAI)[10,11]—and five new indices—the Differential Sector Index (DSI), the Opposite Sector Index (OSI), the Center/Surround Index (CSI), the Irregular Astigmatism Index (IAI), and the Analyzed Area (AA). Except for AA, all the statistical and numerical indices used in the study were derived from the spatial distribution of dioptric power in the videokeratographs.

For the calculation of DSI, OSI, CSI, and IAI, area-corrected power was used in place of the videokeratoscope-derived power. In videokeratography, more samples are commonly collected per unit area from the central cornea than from the periphery. To equalize or unbias the calculation of the indices, each power was multiplied by the area of the cornea from which it was derived, and the sum total was divided by the total corneal area analyzed.

Keratoconus is typified in videokeratographs as an area of significant localized steepening. To detect such features in videokeratographs, DSI, OSI, and CSI were developed. The corneal surface was divided into eight pie-shaped sectors, each subtending 45°. This reference pattern was rotated up to 45° in 32 increments relative to the central axis of the contour map (Fig. 1a) to find the corneal sector with the greatest power. Average power in each sector was calculated from the area-corrected refractive power.

DSI reports the greatest difference in average power between any two sectors, and OSI represents the greatest difference of the average power in opposite sectors (Fig. 1b). CSI is the difference in the average area-corrected corneal power between the central area and an annulus surrounding the central area (Fig. 1c). These three indices help to differentiate among normal corneas, regular astigmatism, peripheral steepening keratoconus, and central steepening keratoconus (Fig. 2).

The IAI describes the short-range semimeridional fluctuation of power distribution. IAI is the average

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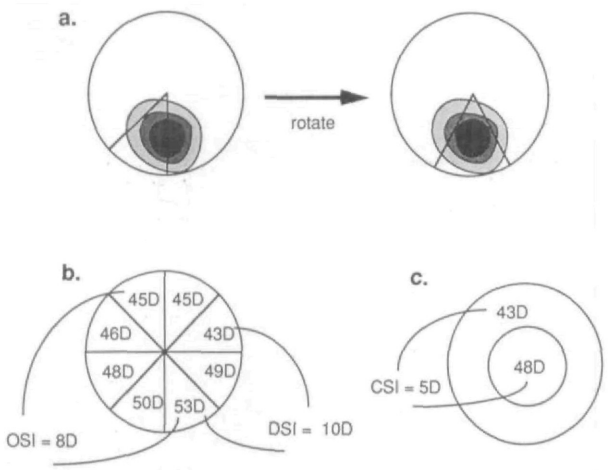

FIGURE 1. Calculation method of Differential Sector Index (DSI), Opposite Sector Index (OSI), and Center/Surround Index (CSI) (**a**) The corneal surface was divided into a pattern of eight pie-shaped sectors, each subtending an angle of 45°. The average power was calculated for each sector based on area-corrected refractive power. New sector regions were defined by rotating the pattern relative to the contour map in 1.41° steps (360°/256 values per mire), up to a maximum of 45°, and calculating average power in each sector at each step. (**b**) DSI reports the greatest difference of the average power between any two sectors obtained during the rotation (53 D − 43 D = 10 D). OSI represents the maximum difference between average powers in opposite sectors obtained during the rotation (53 D − 45 D = 8 D). (**c**) CSI is the difference in the average area-corrected power between the central area (3.0 mm diameter) and an annulus surrounding the central area (3.0 to 6.0 mm) (48 D − 43 D = 5 D).

summation of inter-ring area-corrected power variations along every semimeridian for the entire analyzed surface and is normalized by the average corneal power and number of data points (Fig. 3).

The last index used was the AA, the ratio of the interpolated data area to the area circumscribed by the last mire found in a videokeratoscope image.

Discriminant Analysis Classifier

Two-group discriminant analysis was used as a multivariate statistical technique for the eight indices to screen for keratoconus patterns. A linear discriminant function of the multiple independent variables is found that allows one to discriminate between the two classifications (for example, keratoconus and nonkeratoconus).[12] The linear discriminant function was obtained by discriminant analysis of the training set with a commercially available statistics software package (Statistical Analysis Systems Ver 6.06, SAS Institute, Cary, NC). The linear discriminant function yields a single composite discriminant value for each map, which was designated the Keratoconus Prediction Index (KPI). The division between keratoconus and nonkeratoconus patterns is the cutoff value (Fig. 4). Maps that had a KPI value greater than the optimum cutoff value were classified as keratoconus, whereas maps with a KPI value less than the optimum cutoff value were classified as nonkeratoconus.

The efficacy of the discriminant analysis classifier was tested using the validation set, and the results were described in terms of sensitivity (True positive/[True positive + False negative]), specificity (True negative/[True negative + False positive]), and accuracy ([True positive + True negative]/Total number of maps).

Expert System Classifier

An expert system is a form of artificial intelligence that comprises an extensive set of decision rules. Decisions by the expert system are made deductively with step-by-step logical operation.[13] The discriminant analysis classifier was embedded into a rule-based expert system to enhance keratoconus pattern screening ability and to differentiate between peripheral and central

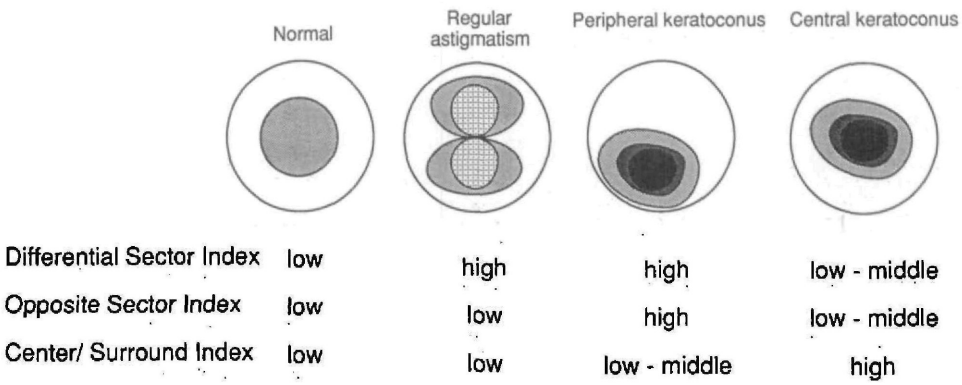

FIGURE 2. The principal differences in pattern of three indices seen in normal, regular astigmatism, and two types of keratoconus. The difference pattern of these three indices is helpful to differentiate normal, regular astigmatism, and keratoconus.

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$$IAI = B * \ln\left[\frac{C * \sum_{i=1,256}^{j=2,30} \{ \Delta A * | P_{i,j} - (P_{i,j+1} + P_{i,j-1})/2 | \}}{\sum_{i=1,256}^{j=2,30} \Delta A} - D\right]$$

i : semimeridional position
j : ring number
$P_{i,j}$: corneal power on the point (i, j)
ΔA : area which corresponds to Power $P_{i,j}$
B : normalization by power
C : normalization by number of points
D : scaling constant

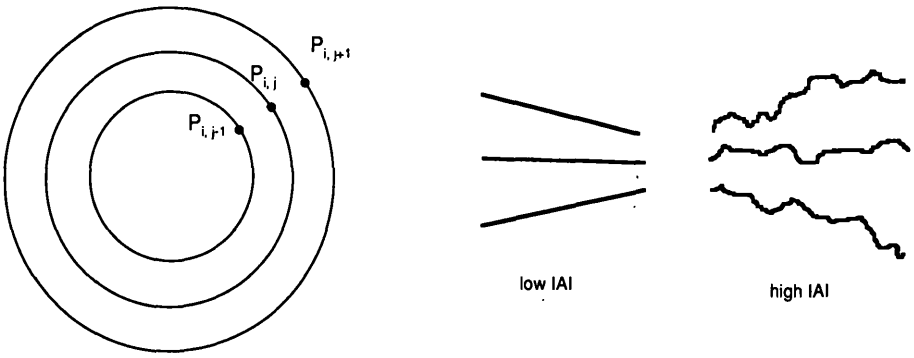

FIGURE 3. Definition of the Irregular Astigmatism Index (IAI). IAI was acquired by calculating the average summation of inter-ring area-corrected power variations along every semimeridian for the entire analyzed surface and normalized by the average corneal power and numbers of all measuring points.

keratoconus patterns. The flow chart of the expert system used in this study is shown in Figure 5 and was implemented with the Pascal program language. In addition to KPI, which was obtained from discriminant analysis, four indices (DSI, OSI, CSI, and SimK2) were used in the binary decision tree. Maps were first classified as either keratoconus, borderline, or nonkeratoconus using KPI and SimK2 values. The borderline maps were then divided into keratoconus or nonkeratoconus by DSI, OSI, and CSI. Next, all keratoconus patterns were classified into either peripheral or central keratoconus using a threshold combination of these indices. Thresholds for DSI, OSI, CSI and SimK2 were developed during the pilot study (unpublished data, 1992). Final output of the system was the display of the certainty of keratoconus relative to the KPI value.

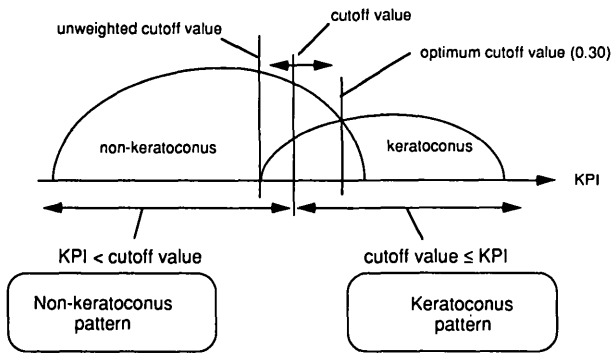

FIGURE 4. The cutoff value in Keratoconus Prediction Index (KPI). The cutoff value is the score against which each KPI is judged to determine into which group the individual map should be classified. The optimum cutoff value is that which permits the least error in classifying the category and was used as the cutoff value in the discriminant analysis. In the expert system classifier, the efficacy of the expert system was evaluated for all cutoff values between the unweighted and the optimum cutoff values.

The most adequate cutoff value was determined from the training set by calculating sensitivity, specificity, and accuracy for several values between the optimum cutoff value and the unweighted cutoff value (Fig. 4). Efficacy of the expert system classifier was evaluated with sensitivity, specificity, and accuracy obtained for the validation set.

RESULTS

Table 2 shows the mean and standard deviation for each of the eight indices used in the keratoconus and nonkeratoconus groups in the training set. Using the unpaired Student's *t*-test, it was possible to show significant differences in six quantitative indices, but reliable case-by-case classification of maps by any single index was not possible because of the large, overlapping ranges. For example, the Opposite Sector Index is similar to the I-S value reported previously.[4] If we set the OSI to distinguish keratoconus at a 95%

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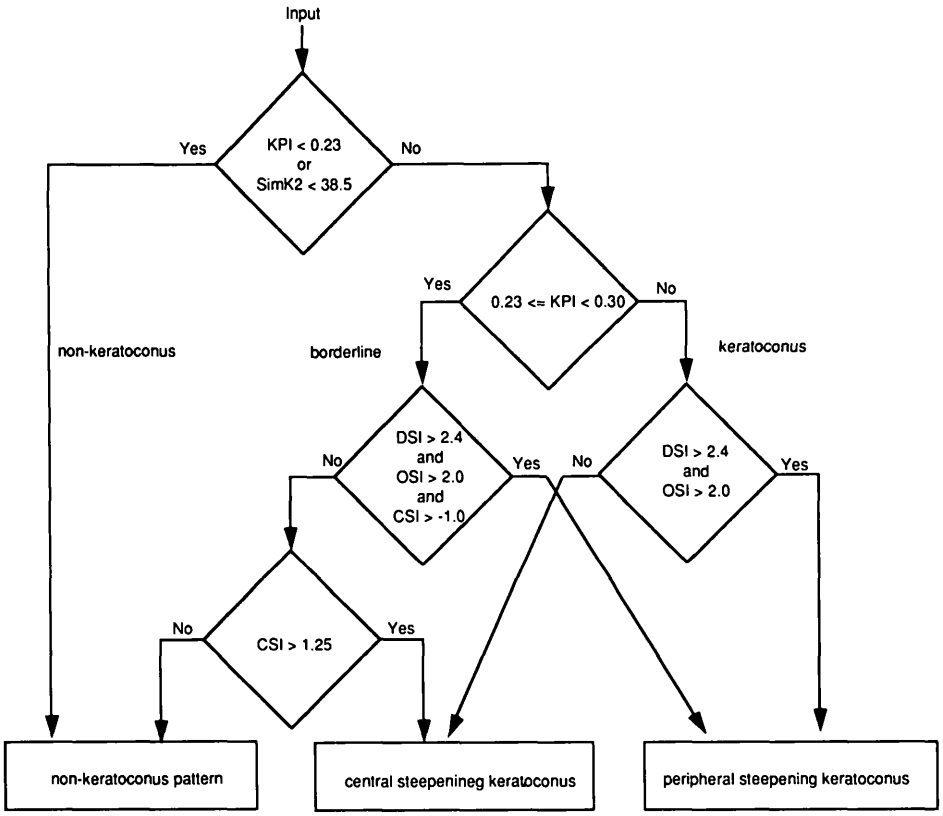

FIGURE 5. Binary decision tree in the expert system. Four indices (DSI, OSI, CSI, SimK2) combined with KPI were used in this decision tree. All cases were finally divided into three groups (nonkeratoconus pattern, peripheral keratoconus pattern, and central keratoconus pattern).

correct level, then there would be 24% false-positive cases in the training set.

Discriminant analysis was performed on the training set and the function obtained was:

$$\mathrm{KPI} = 0.30 + 0.01(-41.23 - 0.15\ \mathrm{DSI} + 1.18\ \mathrm{OSI} + 1.49\ \mathrm{CSI} + 4.13\ \mathrm{SAI} - 0.56\ \mathrm{SimK1} + 1.08\ \mathrm{SimK2} - 3.74\ \mathrm{IAI} + 0.10\ \mathrm{AA}).$$

In this equation, KPI has a value greater than the optimum cutoff value (0.3) when a color-coded map has a keratoconus pattern, and it has a value less than 0.3 when a color-coded map shows nonkeratoconus features. Table 3 shows the mean and standard deviation of KPI for each group in the training and the validation sets. The KPI for keratoconus in each set was significantly higher than the KPIs for any other categories with the unpaired Student's *t*-test.

TABLE 2. Results of All Indices in the Training Set

	Keratoconus (n = 22) (Mean ± SD)	*Nonkeratoconus (n = 78) (Mean ± SD)*	*P Value**
SimK1	53.62 ± 5.72	45.89 ± 4.27	$P = 0.0001$
SimK2	47.54 ± 3.93	42.80 ± 2.87	$P = 0.0001$
Surface Asymmetry Index (SAI)	3.18 ± 2.34	0.60 ± 0.62	$P = 0.0001$
Differential Sector Index (DSI)	11.16 ± 6.50	3.99 ± 3.71	$P = 0.0001$
Opposite Sector Index (OSI)	9.37 ± 5.47	1.84 ± 2.02	$P = 0.0001$
Center/Surrounding Index (CSI)	0.35 ± 3.00	−0.28 ± 0.90	$P = 0.1111$
Irregular Astigmatism Index (IAI)	0.59 ± 0.24	0.47 ± 0.15	$P = 0.0039$
Analyzed Area (AA)	71.79 ± 21.85	78.15 ± 15.16	$P = 0.1203$

* Unpaired Student's *t*-test.

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TABLE 3. Keratoconus Prediction Index (KPI)

	Training Set (Mean ± SD)	*Validation Set (Mean ± SD)*
Keratoconus	0.38 ± 0.07	0.34 ± 0.09
Normal cornea	0.20 ± 0.01†	0.20 ± 0.02†
Keratoplasty	0.20 ± 0.04†	0.17 ± 0.06†
Epikeratophakia	0.19 ± 0.05†	0.17 ± 0.11†
Photorefractive keratectomy	0.18 ± 0.02†	0.20 ± 0.03*
Radial keratotomy	0.17 ± 0.04†	0.16 ± 0.03†
Contact lens-induced warpage	0.21 ± 0.01†	0.19 ± 0.02†
Other	0.20 ± 0.02†	0.19 ± 0.03†

Unpaired *t*-test with keratoconus. *$P < 0.05$; †$P < 0.01$.

Table 4 shows the classification determined by discriminant analysis, compared with the actual diagnosis. Nineteen out of 22 cases of clinically diagnosed keratoconus were detected, with no false-positive cases in the training set. In the validation set, 19 out of 28 keratoconus cases were detected with one false-positive case. Sensitivity, specificity, and accuracy were 86%, 100%, and 97% in the training set, and 68%, 99%, and 90% in the validation set, respectively.

In Table 5, classification predicted by the expert system versus actual classification is shown. Sensitivity, specificity, and accuracy in the training set and in the validation set are shown in Figures 6 and 7, respectively. The optimum cutoff value in the expert system was found to be 0.23 for the most sensitive and accurate results in the training set. Here, sensitivity, specificity, and accuracy were 100%, 96%, and 97% in the training set, and 89%, 99%, and 96% in the validation set, respectively (Table 6).

Using the expert system classifier at a 0.23 KPI cutoff value, all false-positive maps (three in the training set and one in the validation set) were post-keratoplasty eyes. It is important to note that localized abnormal steepening was seen in these maps, and without referring to the patient's history, interpretation of the color-coded map alone would have been similarly inconclusive, even to a well-trained observer. Three false-negative maps in the validation set had typical keratoconus in the contralateral eye. Maps of these false-negative cases showed either a mild inferior steepening, a pellucid marginal degeneration-like pattern, or with-the-rule astigmatism with some irregularity. Thus, it is not surprising that these cases were incorrectly interpreted.

In the validation set, the expert system classifier detected six more keratoconus maps compared to the discriminant analysis classifier alone without increasing false-positive maps. The sensitivity with the expert system classifier (89%) was significantly better than the sensitivity with discriminant analysis alone (68%) (McNemar test with correction for small expected frequencies[14]; $P = 0.016$).

DISCUSSION

Computer-assisted videokeratography and the color-coded map provide an abundance of information about corneal surface characteristics. However, human visual interpretation is essentially subjective, whereas contour information is difficult to analyze quantitatively. An objective assessment of videokeratography is essential for statistical studies of the progression of keratoconus, genetic studies, or screening procedures used for refractive surgery practice. Therefore, the thousands of data points in a color-coded map must be reduced in some fashion to a series of statistically manageable indices.

Rabinowitz and McDonnell[4] reported the first numerical method to differentiate between keratoconus patterns and normals based on videokeratoscopy. They used central corneal power, difference in central corneal power between fellow eyes, and the Inferior-Superior (I-S) value. The I-S value was defined as an average refractive power difference between five inferior points and five superior points 3 mm from the center at 30° intervals. These three parameters were significantly different in patients with keratoconus than in normal controls.

TABLE 4. Classification by Discriminant Analysis Alone

	Predicted Category				
Actual Category	*Keratoconus*	*Nonkeratoconus*	*Sensitivity*	*Specificity*	*Accuracy*
Training set			86%	100%	97%
Keratoconus	19	3			
Nonkeratoconus	0	78			
Validation set			68%	99%	90%
Keratoconus	19	9			
Nonkeratoconus	1	71			

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TABLE 5. Results of Categorization With Expert System

Cutoff Value (KPI)	Group	Training Set KC	Training Set NonKC	Validation Set KC	Validation Set NonKC
0.30*	KC	19	3	19	9
	NonKC	0	78	1	71
0.29	KC	19	3	19	9
	NonKC	0	78	0	72
0.28	KC	19	3	21	7
	NonKC	1	77	0	72
0.27	KC	20	2	21	7
	NonKC	1	77	1	71
0.26	KC	20	2	21	7
	NonKC	2	76	1	71
0.25	KC	21	1	23	5
	NonKC	2	76	1	71
0.24	KC	22	0	23	5
	NonKC	3	75	1	71
0.23	KC	22	0	25	3
	NonKC	3	75	1	71
0.22	KC	22	0	25	3
	NonKC	4	74	1	71
0.21	KC	22	0	25	3
	NonKC	5	73	1	71

KC, keratoconus; NonKC, nonkeratoconus.
* Discriminant analysis only.

However, central corneal power by itself is an inadequate criterion; we have observed families of emmetropes whose central corneal power averages 48 to 50 D. Furthermore, other corneal pathologies (corneal grafts, refractive surgeries, extracapsular cataract extraction, and so on) show asymmetric topographic patterns between the superior and inferior cornea similar to those of keratoconus. In addition, steepening in keratoconus is not limited to the inferior periphery.[5]

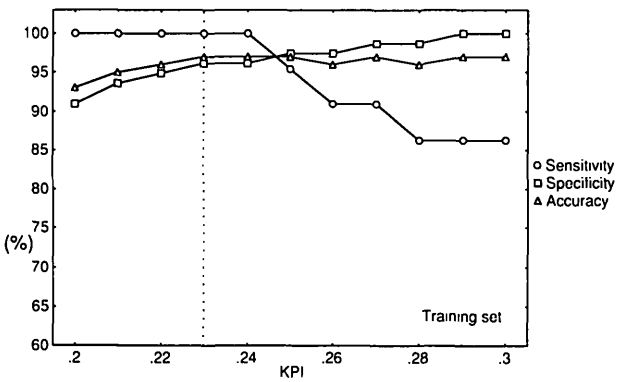

FIGURE 6. Sensitivity, specificity, and accuracy of keratoconus pattern detection in the training set with the expert system. The lower the cutoff value was set, the higher the sensitivity and the lower the specificity. The most efficient cutoff value was found to be 0.23.

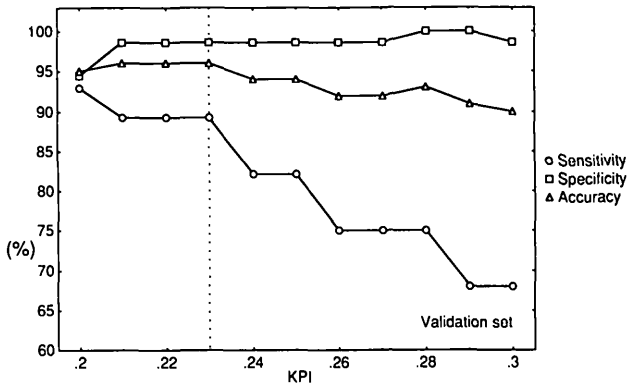

FIGURE 7. Sensitivity, specificity, and accuracy of keratoconus pattern detection in the validation set with the expert system. The lower the cutoff value was set, the higher the sensitivity and the lower the specificity. Note that the cutoff value of 0.23 also showed higher sensitivity with highest accuracy in the validation set.

Comparison between fellow eyes is generally useful for detecting the keratoconus pattern but may be problematic for detecting cone progression because cone development in the fellow eye may actually lag behind the eye in question, or the contralateral cornea may have undergone surgery. Contralateral eye information is not routinely available in videokeratography and should not be relied upon for quantitative analysis. Thus, dioptric maps cannot be reliably and universally distinguished as keratoconus or nonkeratoconus based on the method of Rabinowitz and McDonnell.

To detect topographic characteristics of keratoconus quantitatively, the use of multiple parameters, each of which represents distinctive characteristics of the map, is desirable. Keratoconus patterns in videokeratoscopy can be characterized by an area of localized, abnormal steepening. Localized steepening is often observed in the inferior quadrant, but sometimes it is seen in the center or superior portion of the cornea.[5] This results in asymmetry and a large refractive power difference across the corneal surface. In this study, we used eight indices to extract these characteristics. The DSI and OSI are sensitive to a localized abnormal steepening in the periphery, and the CSI is sensitive to a centrally located steepening (Fig. 2). SAI is also sensitive to topographic asymmetry of decentered cones. IAI and AA describe the characteristic irregularity of the corneal power distribution often associated with moderate to severe keratoconus. SimK1 and SimK2 were used to detect corneal steepening, and the amount of cylinder also associated nonexclusively with keratoconus.

Discriminant analysis was used as a multivariate analysis of these indices. When dependent variables are categorical and the independent variables are di-

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TABLE 6. Classification by Expert System (Cutoff Value 0.23)

	Predicted Category				
Actual Category	*Keratoconus*	*Nonkeratoconus*	*Sensitivity*	*Specificity*	*Accuracy*
Training set			100%	96%	97%
Keratoconus	22	0			
Nonkeratoconus	3	75			
Validation set			89%	99%	96%
Keratoconus	25	3			
Nonkeratoconus	1	71			

mensional, discriminant analysis is considered one of the most appropriate statistical techniques.[12] KPI is the index that is proportional to the discriminant cutoff value obtained from the discriminant function. Average KPI values for keratoconus were significantly higher than those of any other category. This means that KPI is able to differentiate keratoconus not only from normal corneas but also from keratoplasty, epikeratophakia, photorefractive keratectomy, radial keratotomy, and contact lens-induced warpage. Although one can obtain a system that can perform keratoconus screening with high specificity and high accuracy, discriminant analysis by itself is not sufficient for clinical screening of the keratoconus pattern because of relatively low sensitivity (68% in the validation set).

An expert system is an artificial intelligence approach that contains a modifiable knowledge base (rules and facts), an interface to users, and an interface engine that makes logic-based decisions. Expert systems have been applied previously to medical instruments as a tool of computer-assisted diagnosis.[13,15] We created such an expert system through the combination of discriminant analysis with a classification tree to achieve a system for keratoconus pattern screening that was better than the discriminant classifier alone. The current program with the cutoff value 0.23 could correctly differentiate keratoconus and nonkeratoconus maps in 96% of cases. Although three false-negative cases were clinically diagnosed as keratoconus and the contralateral eyes of these patients showed the typical topographic appearance of keratoconus, we were unable to confirm the diagnoses in this retrospective study. In fact, the three false-negative maps resembled contact lens-induced corneal warpage in two eyes and pellucid marginal degeneration in one eye. All the false-positive cases involved eyes that had undergone keratoplasty, and areas of localized, abnormal steepening similar in topographic appearance to keratoconus were seen in these maps. These false-negative and false-positive cases are reasonable under the current conditions in which there is no additional information besides the topographic maps.

In this study, the new methodology enabled the interpretation of topographic patterns quantitatively and objectively, an advantage over the subjective visual inspection of the color-coded map. Because the detection scheme presented here is keyed to detect the keratoconus pattern from videokeratographs, contact lens-induced corneal warpage,[16] post-keratoplasty corneas, or other pathologic corneal conditions can become false positive. In fact, such false positives often cannot be detected with visual inspection of the videokeratograph alone. The clinical diagnosis of keratoconus must be confirmed with more traditional clinical methods. To compare classifiers with well-established tests or findings that unequivocally define the disease, keratoconus suspect patients were excluded because there are no other accepted standards except corneal topography itself. Although they show the keratoconic topographic pattern, some of the keratoconus suspect cases become false negative with the classifier because it was trained intentionally with clinically diagnosed keratoconus.

The ability to screen automatically for keratoconus patterns will be a beneficial tool in the clinician's armamentarium. With experience, the examiner can interpret topographic abnormalities based on the indices and can gain new insight into the relationship between the appearance of the color-coded map and specific measures of corneal power distribution. Additionally, this system can be adapted to detecting other patterns in corneal topography by developing appropriate quantitative indices sensitive to the unique characteristics of those patterns.

Key Words

corneal topography, keratoconus, diagnostic screening, discriminant analysis, expert system

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